D-003-I-001 — HCOS™ Healthcare Systems Instrument

A Human-Collaborative Guide for Understanding and Improving Healthcare Systems

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Document Information

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Issuance Statement

This document is issued as **Version 1.1 — Draft** for public learning, structured reflection, testing, research, feedback, and continued development.

Version 1.1 adds an annotated evidence foundation connecting selected research and authoritative classification guidance to the instrument's design, safeguards, and interpretation boundaries.

The HCOS™ Healthcare Systems Instrument helps people examine healthcare concerns through a human-centered systems lens.

It may be used as:

- A guided reflection tool
- A facilitated conversation structure
- A systems-review worksheet
- A case-analysis framework
- An educational resource
- An AI-enabled collaborative instrument

This instrument does not establish a clinical practice guideline, legal requirement, regulatory standard, accreditation requirement, or independent healthcare decision-making system.

It does not replace:

- Clinical judgment
- Professional expertise
- Patient evaluation
- Emergency guidance
- Legal or regulatory review
- Organizational governance
- Ethical responsibility
- Meaningful human communication
- Accountability for decisions

The instrument supports understanding.

Humans remain responsible for interpretation, verification, judgment, decisions, and action.

Instrument Definition

The HCOS™ Healthcare Systems Instrument is a human-guided tool that helps people examine a healthcare concern by identifying the people affected, the system forces shaping the situation, the pressures being transferred, the protections that may be missing, and the opportunities for responsible improvement.

The instrument may be facilitated by a person, completed as a worksheet, or supported by artificial intelligence.

Regardless of format, it requires meaningful human collaboration.

Evidence Status and Interpretation Boundary

This instrument is **evidence-informed but not yet validated as a complete assessment, diagnostic, predictive, or decision-making instrument**.

The research attached to this document supports selected concepts used within the instrument, including:

- Burnout as an occupational phenomenon associated with chronic workplace stress
- Exhaustion, cynicism or distancing, and reduced professional efficacy as burnout dimensions
- Workload, control, reward, community, fairness, and values as important areas of worklife
- Psychological safety as a condition supporting questions, feedback, help-seeking, error discussion, and team learning
- Civility and the social environment as relevant organizational conditions in healthcare
- The need to distinguish human outcomes from the system conditions that may contribute to them

The attached evidence does **not** establish that:

- The HCOS™ Healthcare Systems Instrument has been clinically or psychometrically validated
- The instrument can diagnose burnout or another health condition
- A specific system force caused an individual outcome
- Every concern identified by the instrument requires the same intervention
- Psychological safety or civility can compensate for inadequate staffing, excessive workload, insufficient authority, missing resources, or inadequate recovery
- An AI-generated reflection is accurate without human verification

Throughout this document, evidence should be interpreted using four categories:

1. **Established finding** — directly supported by the cited source
2. **Author interpretation** — an interpretation made by the source's authors
3. **HCOS™ synthesis** — a reasoned connection between the evidence and the HCOS™ framework
4. **Validation question** — a proposed application that requires future testing

Purpose

The purpose of the HCOS™ Healthcare Systems Instrument is to help people move from:

- A visible problem to a fuller understanding of the system
- Individual blame to responsible systems analysis
- General frustration to identifiable pressure points
- Isolated symptoms to connected causes
- Unstructured concern to a practical next step
- Technology-first thinking to human-centered problem definition
- Assumptions to questions that can be investigated
- Improvement ideas to accountable human action

The instrument helps users ask:

- What is happening?
- Who is affected?
- What are people experiencing?
- What system conditions are shaping the outcome?
- Where is pressure entering the system?
- How is that pressure moving?
- Who is ultimately carrying it?
- Which burdens are necessary?
- Which burdens may be preventable?
- What protections are present or missing?
- What needs to be understood before action is taken?
- What is one responsible next step?

Why This Instrument Exists

Healthcare problems are often first observed through an individual event:

- A patient cannot access treatment
- A referral is delayed
- A clinician misses information
- A family becomes responsible for coordinating care
- A staff member appears disengaged
- A workflow repeatedly fails
- A safety concern is discovered
- A team experiences burnout
- A technology creates unexpected work
- A quality target is not reached
- A policy produces an unintended barrier

The visible event matters.

However, the event may be the final result of multiple interacting system conditions.

For example, a missed follow-up may involve:

- An unclear workflow
- Incomplete documentation
- Competing workload
- A technology limitation
- Unclear ownership
- Inadequate staffing
- A policy that does not match practice
- A metric that prioritizes different work
- A patient-facing communication gap
- Insufficient time for recovery and reflection

When the system is not examined, the final person in the chain may be treated as the entire cause.

That response can create fear without creating understanding.

The HCOS™ Healthcare Systems Instrument exists to help people see the larger system while preserving appropriate individual and organizational accountability.

It does not assume that systems explain every action.

It recognizes that human behavior and healthcare outcomes cannot be understood responsibly without examining the conditions in which they occur.

Core Operating Principle

See the person. See the whole system. Identify where pressure is being carried. Protect human dignity. Improve the conditions rather than asking people to absorb more.

Intended Users

The instrument may be used by:

- Patients
- Family members
- Caregivers
- Healthcare professionals
- Interprofessional care teams

- Healthcare leaders
- Operational teams
- Quality and safety professionals
- Human factors specialists
- Health information technology teams
- Pharmacy teams
- Educators
- Researchers
- Policymakers
- Payers
- Community organizations
- Patient advocates
- AI and technology developers
- Facilitators supporting healthcare improvement

No individual user is expected to understand every part of a healthcare system.

The instrument is strongest when the people affected by the concern are meaningfully included.

Appropriate Uses

The instrument may support examination of:

- Patient access barriers
- Care coordination
- Transitions of care
- Medication-use processes
- Referral pathways
- Scheduling systems
- Documentation
- Communication
- Patient education
- Workforce pressure
- Team collaboration
- Workflow failures
- Technology implementation
- Artificial intelligence implementation
- Policy effects
- Quality measures
- Administrative burden
- Safety concerns
- Patient and caregiver experience
- Organizational decisions
- Resource allocation

- Recovery gaps
- Repeated workarounds
- Unequal healthcare experiences
- Improvement initiatives

It may be used:

- Before designing a solution
- During an improvement initiative
- After an adverse or unexpected event
- When a concern repeatedly returns
- When people disagree about the source of a problem
- Before implementing technology
- When a system appears to depend on individual heroics
- When a policy works on paper but not in practice
- When the people carrying the burden are not being heard

Inappropriate Uses

The instrument should not be used as the sole basis for:

- Diagnosing a patient
- Selecting or prescribing treatment
- Responding to a medical emergency
- Determining immediate patient disposition
- Making an autonomous clinical decision
- Overriding professional judgment
- Determining legal liability
- Conducting a formal disciplinary investigation
- Making an employment decision
- Determining eligibility for healthcare
- Allocating scarce clinical resources
- Replacing informed consent
- Replacing patient or workforce engagement
- Declaring that an individual or organization is safe
- Certifying regulatory compliance
- Reaching a conclusion when essential information is unavailable

The instrument should not be used to create the appearance of participation when decisions have already been made.

It should not be used to shift responsibility from identifiable human decision-makers to an AI system, score, checklist, or generated report.

Immediate Safety and Escalation

The HCOS™ Healthcare Systems Instrument is primarily a reflection and improvement tool.

It is not an emergency-response system.

Before beginning a full systems review, ask:

Is anyone currently at risk of immediate harm?

When there is an urgent patient-safety concern, medical emergency, threat of violence, abuse, neglect, privacy breach, or other immediate danger:

1. Follow the appropriate emergency, clinical, organizational, legal, or regulatory process.
2. Notify the person or team with the authority to respond.
3. Protect the people who may be affected.
4. Preserve relevant information when appropriate.
5. Complete the deeper systems review after the immediate concern has been addressed.

Systems reflection should never delay necessary protection.

Required Human Collaboration

The instrument cannot fully know:

- The complete clinical context
- The history of the organization
- Local professional responsibilities
- The meaning of relationships
- Unwritten workflow realities
- Patient and caregiver values
- The credibility of each source
- Legal and regulatory requirements
- The consequences of a proposed change
- What people may be afraid to say
- Whether the information entered is complete
- How power affects participation
- What responsible action requires in the specific setting

Humans must therefore contribute:

- Context
- Lived experience
- Professional expertise
- Ethical judgment

- Evidence review
- Verification
- Understanding of local conditions
- Awareness of power and relationships
- Authority to make or approve decisions
- Accountability for the results

The instrument may support the analysis.

It does not own the conclusion.

Instrument Inputs

Users should provide as much relevant information as they can responsibly and lawfully share.

Useful inputs may include:

The Concern

- What happened?
- What is happening repeatedly?
- What decision is being considered?
- What feels unsafe, confusing, burdensome, or unsustainable?

The People

- Who receives care?
- Who provides care?
- Who coordinates the process?
- Who makes decisions?
- Who experiences the consequences?
- Whose perspective has not yet been included?

The Setting

- Where does the concern occur?
- When does it occur?
- How often does it occur?
- Is it local, departmental, organizational, or system-wide?

The Existing Process

- What is supposed to happen?
- What actually happens?

- Where are the differences?
- What informal workarounds exist?

Available Evidence

- Policies
- Procedures
- Workflow maps
- Patient or staff feedback
- Safety reports
- Operational data
- Quality measures
- Research
- Technology specifications
- Relevant laws or professional requirements

Known Constraints

- Staffing
- Time
- Financial resources
- Technology
- Regulation
- Physical space
- Training
- Authority
- Information
- Competing priorities

Users should avoid entering protected, confidential, proprietary, or personally identifiable information into an AI-enabled version unless the system has been approved for that information and appropriate safeguards are in place.

Instrument Process

The HCOS™ Healthcare Systems Instrument follows the HCOS™ Journey:

1. Think Well
2. Teach Well
3. Understand Well
4. Grow Well

The 8 + 3 + 1 Framework is used within this journey to examine the system in greater depth.

Step 1 — Define the Concern

Begin with a neutral description of the concern.

Ask:

- What has been observed?
- Who identified the concern?
- What is known?
- What has been assumed?
- Why does this matter?
- What may happen if nothing changes?
- Is this a single event, a recurring pattern, or both?

Avoid beginning with a conclusion about who is at fault.

Helpful Format

Observed situation:

What was seen, heard, measured, or experienced?

Human effect:

How are patients, caregivers, staff, professionals, or communities affected?

System concern:

What process, condition, decision, or relationship may need to be better understood?

Reason for review:

Why is a systems review needed now?

Step 2 — Think Well

See the People

Identify everyone who may be affected.

Consider:

- Patients
- Families
- Caregivers

- Clinicians
- Pharmacists
- Nurses
- Technicians
- Support staff
- Administrative teams
- Leaders
- Community partners
- Payers
- Technology teams
- Future patients or staff

Ask:

- What might each person be experiencing?
- What do they need?
- What matters to them?
- What responsibility do they carry?
- What authority do they have?
- What pressure are they absorbing?
- What may they be afraid to say?
- Whose dignity may be at risk?
- Whose experience is missing?

Protect Dignity

Describe people without reducing them to:

- A diagnosis
- A behavior
- A metric
- A mistake
- A role
- A cost
- A productivity level
- A barrier
- A compliance label

Recognize both agency and context.

People can be responsible for their actions while also being affected by the systems around them.

Preserve What Is Working

Ask:

- What is helping?

- What should not be lost?
- Which relationships are protective?
- Which workarounds reveal valuable human knowledge?
- What strengths already exist?
- Where are people successfully preventing harm?

The goal is not to redesign everything.

It is to understand what should be protected as improvement occurs.

Step 3 — Teach Well

Before people can participate in improvement, they need a shared understanding of the concern.

Ask:

- Is the problem described in understandable language?
- Are technical terms explained?
- Do people understand how the current process works?
- Are roles and responsibilities clear?
- Are the relevant constraints visible?
- Can people distinguish facts from interpretations?
- Has uncertainty been acknowledged?
- Do participants understand what decision is being considered?

Create a Shared Explanation

Explain:

1. What the system is intended to do
2. What appears to be happening
3. Who may be affected
4. What remains uncertain
5. What the review will examine
6. How people can contribute

Teaching should increase participation.

It should not be used to persuade people that a predetermined answer is correct.

Step 4 — Understand Well

Use the eight system forces to examine how the concern developed.

Force 1 — Workload

Examine the amount, intensity, complexity, urgency, and emotional weight of the work.

Ask:

- How much work exists?
- Has all work been counted?
- Who carries it?
- Is the work predictable or constantly changing?
- How much of it is urgent?
- What emotional labor is involved?
- What work occurs after scheduled hours?
- Is demand matched by capacity?
- What is delayed or dropped when workload rises?
- Does the system depend on people doing more than can reasonably be sustained?

Evidence Connection

Maslach and Leiter (2016) identify workload as one of six areas of worklife in which a persistent mismatch between a person and the working environment may contribute to burnout. This supports examining whether demand exceeds sustainable capacity.

This connection should not be used to conclude that workload alone caused burnout in a specific person. Workload must be considered with control, reward, community, fairness, values, resources, workflow, and recovery.

Possible Warning Signs

- Chronic backlog
- Repeated overtime
- Missed breaks
- Excessive task switching
- Unfinished documentation
- Delayed follow-up
- Work completed from home
- Increasing errors or near misses
- Dependence on one highly experienced person
- Emotional exhaustion
- Patient or staff waiting without clear communication

Force 2 — Workflow

Examine how work, information, decisions, and responsibility move.

Ask:

- What is supposed to happen?
- What actually happens?
- Where does the process stop?
- Where is work repeated?
- Where does information fail to transfer?
- Which steps depend on memory?
- Where is ownership unclear?
- Who notices when something is missing?
- Who repairs the process when it fails?
- Are patients or families coordinating the system themselves?
- Which workarounds have become routine?

Possible Warning Signs

- Duplicate entry
- Repeated phone calls
- Manual tracking
- Unclear handoffs
- Missing follow-up
- Multiple sources of truth
- Unnecessary approvals
- Delayed escalation
- Unowned work queues
- Patients repeating the same information
- Informal spreadsheets holding critical processes together

Force 3 — Policies

Examine formal and informal rules.

Ask:

- What policy shapes the concern?
- What purpose was the policy intended to serve?
- Does it work as intended?
- Does written policy match daily practice?
- What work does compliance require?
- Who carries that work?
- Does the policy create an unintended barrier?
- Is there a safe exception or appeal process?
- Can staff report that the policy is not working?
- Has the policy been reviewed since conditions changed?

Possible Warning Signs

- Policies that cannot be followed in practice
- Conflicting requirements

- Rules without clear ownership
- Excessive exception handling
- Unequal effects
- Staff fear of raising concerns
- Policies designed without frontline participation
- Requirements that move burden to patients or families

Force 4 — Metrics

Examine what the system measures and rewards.

Ask:

- What is being measured?
- Why was the measure selected?
- What behavior does it encourage?
- What meaningful work is not measured?
- Can the target be met while the true purpose is missed?
- Does the measure create new administrative work?
- Are patient and workforce effects visible?
- Who reviews unintended consequences?
- Does the measure support learning or create fear?
- Is the data accurate enough to guide action?

Possible Warning Signs

- People working to the metric rather than the purpose
- Unmeasured coordination work
- Productivity expectations disconnected from complexity
- Data used without context
- Targets that compete with patient needs
- Measures that reward speed but overlook quality
- Workforce harm treated as unrelated to performance

Force 5 — Resources

Examine whether people have what they need.

Ask:

- Are staffing and time sufficient?
- Is the necessary information available?
- Is training adequate?
- Does the technology support the work?
- Are medications, equipment, interpretation, transportation, or specialty services available?

- Do people have authority that matches their responsibility?
- Are resources available when and where they are needed?
- How are resources distributed?
- What happens when a resource is unavailable?
- Who compensates for the gap?

Evidence Connection

The six areas of worklife described by Maslach and Leiter (2016) include **control**, which concerns autonomy, discretion, and influence over work. Within this instrument, control is examined partly through questions about whether responsibility is matched by authority, information, time, and access to resources.

This is an HCOS™ mapping rather than a claim that the eight system forces are identical to the six areas of worklife.

Possible Warning Signs

- Responsibility without authority
- Inadequate training
- Staff performing work outside their role
- Poor technology fit
- Delayed access to expertise
- Insufficient interpretation
- Equipment or supply shortages
- Unequal resource distribution
- Teams relying on personal knowledge rather than accessible information

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Force 6 — Leadership Decisions

Examine how priorities and tradeoffs were established.

Ask:

- What decisions shaped the current conditions?
- Who made them?
- Who participated?
- Whose perspective was absent?
- What pressures influenced the decision?
- What tradeoffs were identified?
- Were human effects considered?
- Where were the consequences expected to appear?
- Who is accountable for monitoring the result?
- Can the decision be revised?
- Is the rationale transparent?

Possible Warning Signs

- Decisions made far from the work
- Frontline participation occurring too late
- Unacknowledged tradeoffs
- Responsibility diffused across committees
- Resource cuts without workload reassessment
- Technology selected before workflow analysis
- Leaders receiving incomplete information
- No clear process for revisiting decisions

Force 7 — Daily Operations

Examine how the system is experienced each day.

Ask:

- What do patients and workers actually experience?
- Does practice match the official process?
- Which problems recur?
- What has become normalized?
- Where does the system depend on heroics?
- What happens during evenings, weekends, absences, or high-volume periods?
- How are concerns escalated?
- How quickly are operational problems resolved?
- What knowledge exists at the frontline that leaders may not see?
- Which small changes could create immediate relief?

Evidence Connection

Maslach and Leiter (2016) identify **community**, **fairness**, and **values** as important areas of worklife. Maslach and Leiter (2017) further emphasize the social environment and workplace civility in healthcare.

These sources support examining whether daily operations expose people to conflict, disrespect, isolation, unfair burden distribution, or pressure to act against professional values. They do not imply that relational improvement can substitute for structural correction.

Possible Warning Signs

- Constant workarounds
- Informal escalation
- Different processes depending on who is working
- Unresolved recurring issues
- Excessive meetings without operational change
- Staff protecting the system through personal sacrifice
- Patients receiving inconsistent instructions
- Critical work depending on personal relationships

Force 8 — Recovery Time

Examine whether people can restore capacity after strain.

Ask:

- Are breaks possible in practice?
- Is there sufficient time between shifts?
- Can people disconnect from work?
- Is administrative time protected?
- Is there support after difficult events?
- Are schedules predictable enough for recovery?
- Does the system allow reflection and learning?
- What happens when recovery is repeatedly interrupted?
- Is exhaustion treated as an individual weakness or a system signal?
- Is recovery included in capacity planning?

Evidence Connection

WHO describes burn-out as an occupational phenomenon associated with chronic workplace stress that has not been successfully managed (World Health Organization, 2019). Maslach and Leiter (2016) identify exhaustion as a central dimension of burnout and emphasize the relationship between workers and workplace conditions.

These sources support treating recurring inability to recover as a possible system signal. They do not establish a universal recovery threshold or allow this instrument to diagnose burnout.

Possible Warning Signs

- Missed meals or breaks
- Work following people home
- Persistent after-hours communication
- No decompression after traumatic events
- Chronic vacancies
- Emotional numbness
- Reduced learning and reflection
- People leaving because rest is impossible
- Recovery treated as something employees must arrange on their own

Step 5 — Examine the Three Human Protections

The system forces explain conditions.

The three protections help determine whether people are being safeguarded within those conditions.

Wisdom

Ask:

- What does the evidence show?
- What does professional experience show?
- What does lived experience show?
- What uncertainties remain?
- Does the proposed response make sense in practice?
- What tradeoffs must be acknowledged?
- Are we solving the correct problem?
- What may happen downstream?
- What would change our conclusion?

Wisdom protects the system from acting confidently on incomplete understanding.

Compassion

Ask:

- Who is suffering?
- Who is carrying the greatest burden?
- Which burden is necessary?
- Which burden may be preventable?
- Who may be excluded?
- Are we asking people to adapt to a condition the system should change?
- What would reduce suffering without creating greater harm elsewhere?
- Are people being treated as problems rather than as people experiencing a problem?

Compassion does not remove standards or accountability.

It keeps their human purpose visible.

Presence

Ask:

- Have we listened to the people closest to the concern?
- Can they speak honestly?
- Is anyone afraid of retaliation?
- Are leaders close enough to understand daily reality?
- Have patients and caregivers been included?

- Are we responding to what is happening or to what the process map says should happen?
- Are we willing to remain with the problem long enough to understand it?
- Are we paying attention to information that challenges our preferred solution?

Presence gives the system access to truth.

Evidence Connection

Edmondson (1999) defines team psychological safety as a shared belief that a team is safe for interpersonal risk-taking. In the original study, psychological safety was associated with learning behaviors such as asking questions, seeking feedback, discussing errors, experimenting, and requesting help.

Within HCOS™, this supports examining whether people can speak honestly, admit uncertainty, and raise concerns. The original study did not measure burnout and was not conducted as a healthcare or AI-governance validation study.

Step 6 — Return to the Meaning Anchor

The meaning anchor for HCOS™ is love.

Within healthcare, love may be expressed through:

- Protection of dignity
- Honest communication
- Relief of suffering
- Responsible stewardship
- Respect for professional duty
- Care for patients and families
- Care for the people providing care
- Willingness to confront harmful conditions
- Commitment to learning
- Refusal to treat people as disposable

Ask:

- Who is this system meant to serve?
- What must be protected?
- What would responsible care require?
- Are we preserving the humanity of the people involved?
- Does the proposed response help people flourish?
- Are we acting from fear, control, convenience, or care?

Love does not mean avoiding difficult decisions.

It means making them without forgetting the human beings who will live with the consequences.

Step 7 — Map the Pressure Pathway

Describe how pressure enters and moves through the system.

Pressure Pathway Template

Pressure source:

Where does the demand, constraint, risk, or requirement begin?

Transfer point:

How is the pressure moved to another person, team, or process?

Human carrier:

Who ultimately absorbs the pressure?

Visible effect:

What does the system observe?

Hidden effect:

What may remain unseen?

Current protection:

What helps prevent harm?

Missing protection:

What safeguard, resource, authority, information, or recovery is absent?

Possible intervention point:

Where could the pathway be interrupted or redesigned?

Key Question

Is the system solving the problem, or is a person absorbing it?

Step 8 — Separate What Is Known From What Is Assumed

Create four categories.

Known

Information supported by reliable observation, documentation, evidence, or direct experience.

Interpreted

A reasonable explanation that has not yet been fully confirmed.

Assumed

A belief that may be influencing the review but has not been adequately examined.

Missing

Information, perspectives, or evidence still needed.

This step helps prevent an AI-generated explanation, leadership assumption, isolated story, or incomplete metric from being treated as established truth.

Step 9 — Identify Improvement Options

Generate possible responses at more than one system level.

Person-Level Support

What immediate support may help the people currently affected?

Workflow-Level Change

What process could be clarified, simplified, removed, reassigned, or redesigned?

Team-Level Change

What communication, role, authority, or collaboration change may help?

Organizational Change

What policy, resource, metric, leadership decision, or governance process should be reconsidered?

Ecosystem-Level Change

What payer, regulatory, vendor, community, educational, or public-policy condition contributes to the concern?

For each option, ask:

- What problem would this address?
- Who would benefit?
- Who might carry new work?
- What risks could it create?
- What evidence supports it?
- What authority is required?
- How would we know whether it helped?
- Can it be reversed if harm appears?
- Does it reduce burden or merely move it?

Step 10 — Grow Well

Select a responsible next step.

The next step should be:

- Human-centered
- Specific
- Proportionate
- Evidence-informed
- Realistic
- Assigned to an identifiable owner
- Supported with adequate resources
- Open to revision
- Measurable without reducing people to a metric

Action Plan

Action:

What will be done?

Purpose:

What human or system need will it address?

Owner:

Who is responsible?

Participants:

Who needs to be included?

Resources:

What is needed?

Timeframe:

When will the action occur?

Protection:

How will patients, staff, privacy, safety, and dignity be protected?

Evaluation:

What evidence will show whether the change helped?

Unintended consequences:

What will be monitored?

Stopping or revision condition:

When should the action be paused, changed, or ended?

Step 11 — Human Review and Accountability

Before acting on the instrument's output, a qualified person or appropriately constituted group should review:

- Accuracy
- Completeness

- Evidence
- Assumptions
- Missing voices
- Equity
- Privacy
- Safety
- Legal and regulatory implications
- Professional responsibilities
- Resource requirements
- Unintended consequences
- Ownership
- Escalation
- Monitoring

The final record should identify:

- Who reviewed the output
- Who approved the action
- Who is accountable for implementation
- How affected people can raise concerns
- When the decision will be revisited

An AI system should never be listed as the accountable decision-maker.

Instrument Output

The standard output should be titled:

HCOS™ Healthcare Systems Reflection Record

The record should include:

1. Concern reviewed
2. Reason for review
3. People affected
4. Immediate safety considerations
5. What is working and should be preserved
6. Workload findings
7. Workflow findings
8. Policy findings
9. Metric findings
10. Resource findings
11. Leadership-decision findings
12. Daily-operations findings
13. Recovery-time findings

14. Wisdom considerations
15. Compassion considerations
16. Presence considerations
17. Meaning-anchor reflection
18. Pressure pathway
19. Known information
20. Interpretations
21. Assumptions
22. Missing information
23. Improvement options
24. Recommended next step
25. Human reviewer
26. Accountable owner
27. Evaluation plan
28. Unintended consequences to monitor
29. Review date
30. Remaining questions

The output is a structured reflection record.

It is not a diagnosis, verdict, certification, or independent decision.

Recommended Response Style

When the instrument is AI-enabled, responses should be:

- Calm
- Respectful
- Clear
- Nonjudgmental
- Evidence-aware
- Honest about uncertainty
- Easy to understand
- Focused on systems rather than blame
- Protective of dignity
- Practical without becoming overly directive

The instrument should:

- Recognize strengths before identifying problems
- Avoid shaming individuals
- Avoid claiming certainty from incomplete information
- Explain difficult concepts simply
- Identify missing information
- Distinguish observation from interpretation
- Avoid overwhelming the user

- Prioritize one meaningful next step when appropriate

User Instructions

How to Use the Instrument

Describe a healthcare concern in your own words.

Helpful information may include:

- What happened
- Who was involved
- Who was affected
- What was expected to happen
- What happened instead
- Whether the concern is recurring
- What has already been tried
- What information is still missing
- What decision or improvement is being considered

Do not include confidential or identifiable patient, employee, or organizational information unless you are using an appropriately approved and protected system.

The instrument will help you:

1. Clarify the concern
2. Identify the people affected
3. Examine the eight system forces
4. Identify pressure pathways
5. Consider the three human protections
6. Separate facts from assumptions
7. Identify areas for deeper review
8. Develop a responsible next step

Review every output carefully.

Correct anything that is inaccurate.

Add context the instrument could not know.

Include the people affected before making decisions.

Facilitator Guidance

A facilitator should:

- Explain the instrument's purpose
- Clarify that it is not a blame exercise
- Establish psychological safety
- Protect confidentiality
- Include affected people
- Prevent one role or level of authority from dominating
- Distinguish evidence from interpretation
- Make uncertainty visible
- Watch for power differences
- Keep the discussion connected to the human experience
- Record disagreements rather than forcing false consensus
- Identify who owns follow-up
- Return findings to participants when appropriate

A facilitator should not use the instrument to pressure participants into accepting a predetermined conclusion.

Evidence-Informed Facilitation Boundary

The instruction to establish psychological safety is informed by Edmondson's (1999) work on interpersonal risk-taking and team learning. The emphasis on civility is also consistent with Maslach and Leiter's (2017) discussion of the healthcare social environment.

A facilitator should not declare a group psychologically safe solely because respectful language is used or participants are invited to speak. Meaningful safety also depends on power, confidentiality, response to concerns, freedom from retaliation, and whether the organization is willing and able to act.

AI-Enabled Instrument Configuration

The following prompt may be used to configure an AI-enabled version of the instrument.

Core Instrument Prompt

You are the **HCOS™ Healthcare Systems Instrument**, a human-collaborative tool for understanding healthcare systems.

Your purpose is to help users examine healthcare concerns while protecting human dignity, identifying system conditions, revealing pressure pathways, and supporting responsible improvement.

You are not an autonomous decision-maker.

You do not replace clinicians, healthcare professionals, patients, caregivers, leaders, legal counsel, regulators, ethics committees, safety teams, or emergency services.

You do not diagnose, prescribe, determine legal liability, certify compliance, or independently decide what action should be taken.

Your role is to support structured human reflection.

Follow the HCOS™ Journey:

1. Think Well
2. Teach Well
3. Understand Well
4. Grow Well

Use the HCOS™ 8 + 3 + 1 Framework.

Examine the eight system forces:

1. Workload
2. Workflow
3. Policies
4. Metrics
5. Resources
6. Leadership Decisions
7. Daily Operations
8. Recovery Time

Examine the three human protections:

1. Wisdom
2. Compassion
3. Presence

Return to the meaning anchor:

- Love of people
- Love of the work
- Protection of dignity
- Relief of unnecessary suffering
- Responsible stewardship

Begin by determining whether anyone may be at immediate risk of harm.

When an urgent safety, medical, legal, privacy, abuse, or emergency concern may exist, clearly direct the user to the appropriate immediate human or organizational response before continuing with systems reflection.

When conducting the review:

- See the people before evaluating the process.
- Recognize strengths and protections that already exist.
- Do not shame or blame.
- Do not assume the visible person is the entire cause.
- Do not remove appropriate individual accountability.
- Distinguish observations, interpretations, assumptions, and missing information.
- State uncertainty honestly.
- Ask where pressure enters, how it moves, and who carries it.
- Ask whether a burden is necessary or preventable.
- Identify where the system depends on human sacrifice or informal workarounds.
- Examine whether a proposed solution reduces burden or transfers it.
- Do not recommend technology until the human problem and existing workflow are understood.
- Do not present AI-generated conclusions as verified facts.
- Keep human judgment, responsibility, and accountability visible.
- Use clear, accessible language.
- Avoid overwhelming the user.
- Identify one practical next step when enough information is available.

Structure the final reflection as:

1. Concern
2. People affected
3. Immediate safety considerations
4. What is working
5. Eight system forces
6. Three human protections
7. Meaning anchor
8. Pressure pathway
9. Known information
10. Interpretations
11. Assumptions
12. Missing information
13. Improvement options
14. Responsible next step
15. Human review required
16. Remaining questions

End with this reminder:

This instrument supports human understanding. People with the appropriate knowledge, responsibility, authority, and relationship to the situation must verify the information and determine what happens next.

Limitations

The instrument may be limited by:

- Incomplete information
- Inaccurate information
- Missing perspectives
- User assumptions
- Organizational power differences
- Limited evidence
- Local conditions
- Unwritten practices
- AI bias
- Model errors
- Outdated information
- Lack of access to relevant policies or data
- Differences among healthcare settings
- Failure to include affected people
- Use by individuals without appropriate authority or expertise

A well-structured analysis can still be wrong.

The instrument's organization and clarity should not be mistaken for proof that its conclusions are accurate.

Safeguards

Human Oversight

All conclusions and actions require human review.

Privacy

Do not enter protected or confidential information into an unapproved system.

Evidence

Verify factual claims and identify the source of evidence used.

Uncertainty

State what is unknown or contested.

Bias

Examine whether the instrument, data, participants, or process may overlook or disadvantage particular people.

Inclusion

Include the people who experience the system whenever possible.

Accountability

Assign decisions and actions to identifiable humans or governance bodies.

Reversibility

When possible, use changes that can be paused or revised if harm appears.

Monitoring

Evaluate intended and unintended effects.

Escalation

Direct urgent concerns to appropriate human authorities.

Burden Review

Ensure that proposed improvements do not quietly create additional work for patients, families, or frontline teams.

Evidence and Development

The instrument should continue to be informed by research and practice involving:

- Human factors
- Systems engineering
- Patient safety
- Implementation science
- Quality improvement
- Organizational behavior
- Psychological safety
- Healthcare workforce well-being
- Burnout and occupational health
- Patient and caregiver experience

- Health literacy
- Shared decision-making
- Health equity
- Ethics
- Leadership
- Healthcare operations
- Technology implementation
- Artificial intelligence governance

Supporting evidence should be documented through the HCOS™ Annotated Evidence Repository.

Evidence should be used not only to support the instrument, but also to challenge and improve it.

Evidence Traceability Requirements

When evidence is used within this instrument, the record should identify:

- The exact source
- The source type
- The population and setting studied
- What the source directly found
- What the source did not study
- How HCOS™ is interpreting or extending the finding
- Whether the application has been validated in the intended healthcare setting
- Any copyright, licensing, or use restrictions associated with a measurement instrument

The presence of research support for an individual concept does not validate the complete HCOS™ Healthcare Systems Instrument.

Review and Revision

Feedback should be collected from:

- Patients
- Caregivers
- Healthcare professionals
- Operational staff
- Healthcare leaders
- Quality and safety teams
- Human factors specialists
- Researchers
- Educators
- Technology professionals

- Community representatives

Future versions should evaluate:

- Clarity
- Accessibility
- Usability
- Reliability
- Human burden
- Bias
- Safety
- Practical value
- Appropriateness across settings
- Whether the instrument improves understanding
- Whether resulting actions improve system conditions
- Whether unintended harm occurs

Material revisions should be recorded in the version history.

Appendix A — Annotated Evidence Foundation

Purpose of This Appendix

This appendix records the principal evidence sources currently connected to the design of D-003-I-001 — HCOS™ Healthcare Systems Instrument.

A source is included because it informs a concept, question, safeguard, or interpretation boundary. Inclusion does **not** mean that the source validates the complete instrument, establishes clinical effectiveness, or authorizes independent decision-making.

Each annotation distinguishes:

- The source's direct contribution
- Its relevance to this instrument
- Its principal evidence boundary

Evidence-to-Instrument Traceability

Evidence source	Direct contribution	Instrument connections	Principal boundary
World Health Organization (2019)	Defines burn-out as an occupational phenomenon associated with chronic workplace stress that has not been successfully managed	Immediate non-diagnostic boundary; workload; recovery time; human-load interpretation	Authoritative classification statement, not a causal, intervention, or instrument-validation study
Maslach and Jackson (1981)	Develops a multidimensional measure of experienced burnout	Recognition of exhaustion, distancing, and reduced accomplishment as human outcomes	A measurement study; does not identify system causes or provide a clinical diagnosis
Maslach and Leiter (2016)	Synthesizes burnout as exhaustion, cynicism, and reduced efficacy and identifies six areas of worklife	Workload; resources and authority; daily operations; recovery; pressure pathways; system redesign	Narrative review; does not prove that a particular mismatch caused an individual outcome
Edmondson (1999)	Defines team psychological safety and links it with learning behavior	Presence; facilitator guidance; help-seeking; error discussion; escalation; learning	Original study did not directly examine burnout, healthcare delivery, or AI governance
Maslach and Leiter (2017)	Reviews civility, the healthcare social environment, and organizational approaches to burnout	Compassion; presence; communication; community; daily operations; facilitator conduct	Civility should complement rather than replace structural reform

AE-D003-I001-001 — Burn-out as an Occupational Phenomenon

Citation

World Health Organization. (2019, May 28). *Burn-out an “occupational phenomenon”: International Classification of Diseases*. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases

Source Type

Authoritative international classification statement.

Direct Contribution

WHO describes burn-out in ICD-11 as a syndrome conceptualized as resulting from chronic workplace stress that has not been successfully managed. It identifies three dimensions:

1. Energy depletion or exhaustion
2. Increased mental distance, negativism, or cynicism related to work
3. Reduced professional efficacy

WHO classifies burn-out as an occupational phenomenon rather than a medical condition and limits the construct to the occupational context. WHO also clarifies that burn-out was included in ICD-10 within the same broad category, while ICD-11 provides a more detailed definition.

Relevance to D-003-I-001

This source supports:

- Treating chronic occupational stress as a legitimate systems-review concern
- Examining workload, resources, leadership decisions, operations, and recovery
- Maintaining a clear boundary between systems reflection and medical diagnosis
- Avoiding the assumption that “not successfully managed” necessarily represents individual failure

Evidence Boundary

The WHO classification does not identify the cause of burn-out in a particular person, prescribe a measurement instrument, establish numerical thresholds, or prove that a particular workplace intervention will be effective. It also does not exclude coexisting medical or mental health conditions requiring appropriate assessment.

AE-D003-I001-002 — Measurement of Experienced Burnout

Citation

Maslach, C., & Jackson, S. E. (1981). The measurement of experienced burnout. *Journal of Occupational Behaviour, 2*(2), 99–113. https://doi.org/10.1002/job.4030020205

Source Type

Psychometric instrument-development study.

Direct Contribution

Maslach and Jackson developed and evaluated a multidimensional measure of experienced burnout. The study supported three related dimensions commonly described as emotional exhaustion, depersonalization, and personal accomplishment.

Relevance to D-003-I-001

This study helps separate:

- **Human outcomes**, such as exhaustion, distancing, or diminished accomplishment
- **Possible system contributors**, such as workload, low authority, poor workflow, conflict, inadequate resources, and missing recovery

Within HCOS™, these dimensions may prompt further systems investigation. They should not be treated as automatic proof of a particular cause.

Evidence Boundary

The Maslach Burnout Inventory does not independently diagnose a medical or psychiatric condition, identify which workplace condition caused a score, or determine the correct intervention. Proprietary inventory items and scoring instructions should not be reproduced or embedded without appropriate authorization and licensing.

AE-D003-I001-003 — Burnout and the Six Areas of Worklife

Citation

Maslach, C., & Leiter, M. P. (2016). Understanding the burnout experience: Recent research and its implications for psychiatry. *World Psychiatry, 15*(2), 103–111. https://doi.org/10.1002/wps.20311

Source Type

Narrative review and conceptual synthesis.

Direct Contribution

Maslach and Leiter describe burnout as an occupational syndrome involving exhaustion, cynicism, and reduced professional efficacy. They identify six areas of worklife in which persistent mismatches may contribute to burnout:

1. Workload
2. Control
3. Reward
4. Community
5. Fairness
6. Values

They emphasize changing workplace conditions rather than relying exclusively on individual coping strategies.

Relevance to D-003-I-001

The six areas inform several parts of this instrument:

Area of worklife	D-003-I-001 connection
Workload	Force 1 — Workload
Control	Resources, authority, leadership decisions, and participation
Reward	Metrics, recognition, purpose, and reciprocity
Community	Daily operations, communication, compassion, and presence
Fairness	Policies, metrics, leadership decisions, and burden distribution
Values	Meaning anchor, professional duty, ethical conflict, and stewardship

This mapping is an HCOS™ synthesis. The six areas of worklife and the eight HCOS™ system forces are related but not identical frameworks.

Evidence Boundary

The article does not establish that all six mismatches must be present, that one mismatch caused burnout in a specific person, or that individual support has no value. Workplace redesign and person-level support may both be necessary.

AE-D003-I001-004 — Psychological Safety and Learning Behavior

Citation

Edmondson, A. C. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly, 44*(2), 350–383. https://doi.org/10.2307/2666999

Source Type

Multimethod organizational field study.

Direct Contribution

Edmondson defines team psychological safety as a shared belief that the team is safe for interpersonal risk-taking. Psychological safety was associated with learning behaviors such as asking questions, seeking feedback, discussing errors, experimenting, and requesting help.

Relevance to D-003-I-001

This study supports instrument features that ask whether people can:

- Admit uncertainty
- Report mistakes and near misses
- Request assistance
- Challenge assumptions
- Raise concerns without retaliation
- Participate in learning and improvement

It also supports facilitator practices that reduce interpersonal risk and preserve disagreement rather than forcing false consensus.

Evidence Boundary

The original study did not measure burnout and was not conducted as a healthcare, patient-safety, or AI-governance validation study. Applying the concept to clinical settings, workforce well-being, or AI oversight requires context-specific evaluation.

Psychological safety cannot compensate indefinitely for inadequate staffing, excessive workload, missing resources, limited authority, or organizational failure to respond.

AE-D003-I001-005 — Civility, Social Environment, and Burnout

Citation

Maslach, C., & Leiter, M. P. (2017). New insights into burnout and health care: Strategies for improving civility and alleviating burnout. *Medical Teacher, 39*(2), 160–163. https://doi.org/10.1080/0142159X.2016.1248918

Source Type

Narrative review and conceptual overview focused on healthcare and medical education.

Direct Contribution

Maslach and Leiter emphasize the importance of the social environment in healthcare and discuss workplace civility as a modifiable organizational condition. The article considers how respectful social encounters and civility-focused interventions may contribute to reductions in burnout.

Relevance to D-003-I-001

This source supports examining relational human load, including:

- Interpersonal vigilance
- Conflict-management burden
- Fear of humiliation or retaliation
- Isolation
- Reluctance to ask for help
- Reduced trust and belonging

It also supports the instrument's emphasis on compassion, presence, respectful facilitation, and protection of dignity.

Evidence Boundary

Civility should not be treated as a substitute for adequate staffing, manageable workload, fair procedures, appropriate authority, sufficient resources, or recovery time. Respectful interaction can reduce relational harm, but it cannot make structurally unsafe work sustainable.

Cross-Evidence HCOS™ Synthesis

Together, these sources support the following design logic:

1. Observe the Human Outcome Without Diagnosing

Exhaustion, distancing, cynicism, and diminished efficacy may be important signals, but this instrument does not diagnose burnout.

2. Investigate the Work System

Potential domains include workload, control, reward, community, fairness, values, workflow, policies, metrics, resources, leadership decisions, daily operations, and recovery.

3. Trace Pressure Rather Than Assigning Immediate Blame

The visible person may be carrying pressure created elsewhere in the system. This possibility should be investigated without removing appropriate individual accountability.

4. Protect Voice and Learning

People need conditions in which they can ask for help, discuss errors, report pressure, and challenge unsafe assumptions.

5. Address Relational and Structural Conditions Together

Psychological safety and civility are meaningful protections. They should be accompanied by action on workload, staffing, authority, resource, policy, workflow, and recovery problems.

6. Distinguish Evidence From HCOS™ Interpretation

Outputs should identify whether a statement represents:

- A direct research finding
- An author interpretation
- An HCOS™ synthesis
- A hypothesis requiring testing
- A proposed application
- An unresolved question

7. Validate the Instrument Itself

Future research should examine whether use of D-003-I-001 improves:

- Problem definition
- Inclusion of affected perspectives
- Identification of system conditions
- Recognition of transferred human burden
- Quality of improvement options
- Psychological safety during review
- Clarity of accountability
- Detection of unintended consequences

It should also evaluate burden, reliability, usability, bias, safety, and outcomes across different healthcare settings.

Appendix B — Bibliography

Edmondson, A. C. (1999). Psychological safety and learning behavior in work teams. *Administrative Science Quarterly, 44*(2), 350–383. https://doi.org/10.2307/2666999

Maslach, C., & Jackson, S. E. (1981). The measurement of experienced burnout. *Journal of Occupational Behaviour, 2*(2), 99–113. https://doi.org/10.1002/job.4030020205

Maslach, C., & Leiter, M. P. (2016). Understanding the burnout experience: Recent research and its implications for psychiatry. *World Psychiatry, 15*(2), 103–111. https://doi.org/10.1002/wps.20311

Maslach, C., & Leiter, M. P. (2017). New insights into burnout and health care: Strategies for improving civility and alleviating burnout. *Medical Teacher, 39*(2), 160–163. https://doi.org/10.1080/0142159X.2016.1248918

World Health Organization. (2019, May 28). *Burn-out an "occupational phenomenon": International Classification of Diseases*. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases

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Repository Structure

```
D-003-Healthcare-Systems/
└── 08-instruments/
    └── healthcare-systems-instrument/
        ├── README.md
        ├── user-guide.md
        ├── facilitator-guide.md
        ├── instrument-prompt.md
        ├── reflection-record-template.md
        ├── examples/
        ├── research/
        └── version-history.md
```

For the initial release, the complete instrument may remain in `README.md`.

The supporting files can be separated as the instrument develops.

Version History

Version	Date	Status	Description
1.1	July 19, 2026	Draft	Added annotated evidence connections, evidence-status and interpretation boundaries, evidence-to-instrument traceability, focused evidence notes, research questions, and bibliography.
1.0	July 19, 2026	Draft	Initial issuance of the HCOS™ Healthcare Systems Instrument, including its purpose, human-collaboration model, review process, safeguards, output structure, and AI-enabled configuration.

Suggested Citation

Sophia Care Alliance. *D-003-I-001 — HCOS™ Healthcare Systems Instrument: A Human-Collaborative Guide for Understanding and Improving Healthcare Systems.* Version 1.1—Draft. Issued July 19, 2026. Human-Centered Operating Systems™; 2026.

Closing Principle

A healthcare problem should not be reduced to the last person who touched it.

The HCOS™ Healthcare Systems Instrument helps people see the person, understand the conditions, trace the pressure, protect what matters, and identify a healthier next step.

The instrument supports the work.

Humans provide the context, evidence, judgment, compassion, authority, and accountability.

Think Well.

Teach Well.

Understand Well.

Grow Well.

Lead with Love.

See the Whole System.

Protect with Wisdom, Compassion, and Presence.

Help People Flourish.